CS217 - Data Structures & Algorithm Analysis (DSAA)

Lecture #9

Binary Search Trees

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Reading: Chapter 12

Aims of this lecture

- We've seen a lot of binary trees already
  - Recurrence tree for visualising runtime in recursive calls
  - HeapSort uses imaginary trees
  - Decision trees in the lower bound for comparison sorts
- Now: discussing binary trees more thoroughly, including how to prove **inductive statements about trees**.
- To introduce **binary search trees** and their typical operations.
- To work out the **running time** for operations on binary search trees.

Recall

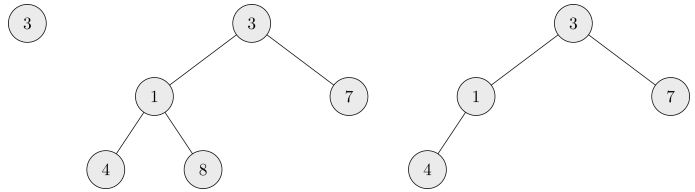
- Elements can contain **satellite data** and a **key** is used to identify the element.
- Typical operations:
  - **Search(S, k):** returns element x with **key k**, or NIL
  - **Insert(S, x):** adds **element x** to S
  - **Delete(S, x):** removes element x from S
  - **Minimum(S), Maximum(S):** return x resp. with smallest or largest key
  - **Successor(S, x), Predecessor(S, x):** next larger (smaller) than Key(x)
- **Time** often measured using n as the number of elements in S.

Binary trees

- Intuitively: trees where every node has at most two children.
- We can define binary trees recursively:
- A **binary tree** is a structure defined as finite set of nodes such that either
  - The tree is empty (no nodes) or
  - It is composed of a root node, a left subtree and a right subtree
- This view is very handy for proving statements about trees by induction (see later).
- The root of the left subtree of a node is called **left child**, that of the right subtree is called **right child**.

Definitions for binary trees

- We tacitly assume that all nodes are labelled by numbers.



- A **path** in a tree is a 不重复的 sequence of nodes linked by edges. The **length** of a path is the number of edges.
- A **leaf** of a tree is a node that has no children; otherwise it is called **internal node**.
- We speak about **siblings**, **parents**, **ancestor**, **descendant** in the obvious way.

Depth and height

- The **depth of a node** in a tree is the length of a (simple) path from that node to the root.
- A **level** of a tree is a set of nodes of the same depth.
- The **height of a node** in a tree is the length of the longest path from that node to a leaf.
- The **height of a tree** is the height of its root.
- A binary tree is **full** if each node is either a leaf or has exactly two children.
- It is **complete** if it is full and all leaves have the same level.

Inductive proofs on trees

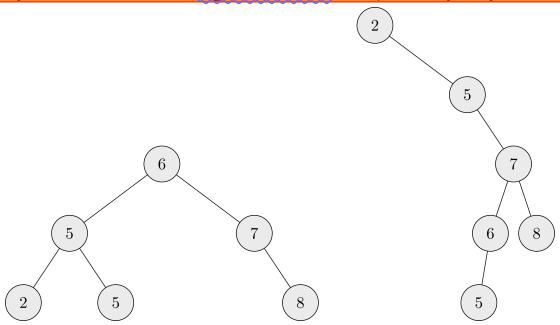
- We can use the recursive definition to prove statements about trees inductively. The general recipe is this:
- **Proof**:
  - **Base case**: show that the statement holds for the "smallest" tree, e.g. an empty tree or just the root node (depending on the statement).
  - **Induction step**: any larger tree has a root and two subtrees (possibly empty). Assume that the statement holds for both subtrees and show that it then holds for the whole tree.
- Caveat: if a statement reads "for all non-empty trees", in the induction step we may need to watch out for empty subtrees.

Inductive proofs on trees: example

- **Theorem**: A binary tree of height at most h has no more than 2^h leaves.
  - We have used this statement in the lower bound for comparison sorts. Now we prove it.
- **Proof**:
  - **Base case**: a tree of height 0 has no more than $2^0=1$ leaves.
  - **Induction step**: a tree of height $h>0$ has a root and two subtrees (possibly empty) of height at most $h-1$. Assume that the statement holds for both subtrees. Then the subtrees have at most 2^{h-1} leaves, so the whole tree has at most $2\cdot 2^{h-1}=2^h$ leaves.

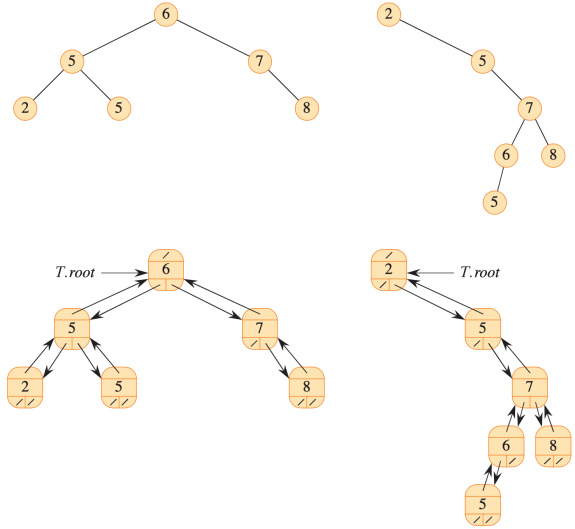
Binary search trees

- A binary search tree (BST) is a binary tree where all labels (keys) satisfy the **binary search tree property**:
  - If y is a node in the left subtree of x, then y.key ≤ x.key.
  - If y is a node in the right subtree of x, then y.key ≥ x.key.



Binary search trees: representation

- Linked list
- Key
- Satellite data
- Attributes:
  - **T.Root**
  - **Left** child pointer
  - **Right** child pointer
  - **Parent** pointer
- Parent of T.root is NIL

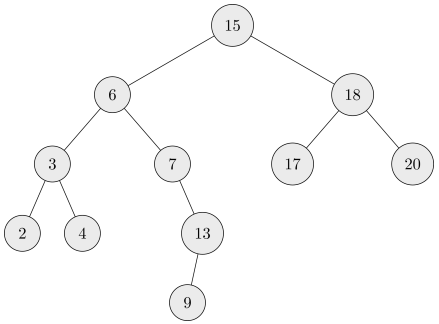


返回 NIL 正是这个函数用来“发出信号”的方式，告诉调用者：“你要找的那个键 k 在这棵树里不存在。”

Searching in a BST

- **Search(x, k):** returns the element with key k in a tree rooted in x, or NIL
- **Idea**: compare against current key and stop or go down left or right.

找不到了 (Failure Base Case): x == NIL
这表示我们已经“掉出了”树。



```
TREE-SEARCH(x, k)
if x == NIL or k == x.key
    return x
if k < x.key
    return TREE-SEARCH(x.left, k)
else return TREE-SEARCH(x.right, k)

ITERATIVE-TREE-SEARCH(x, k)
while x ≠ NIL and k ≠ x.key
    if k < x.key
        x = x.left
    else x = x.right
return x
```

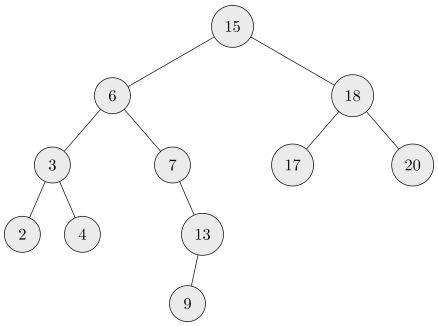
Runtime: *O(h), h* the height of the tree

NIL is a sign for termination, which means we EITHER go to the right extreme pt and both childs are leaves which have value NIL, OR we cannot find the pt and we end up with the NIL of the whole mother tree.

Minimum, Maximum, Successor in a BST

Minimum: starting from the root, go left until the left child is NIL.

Maximum: starting from the root, go right until the right child is NIL.



```
TREE-MINIMUM(x)
while x.left ≠ NIL
    x = x.left
return x

TREE-MAXIMUM(x)
while x.right ≠ NIL
    x = x.right
return x
```

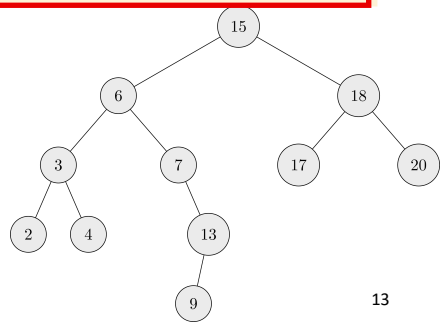
Runtime: *O(h), h* the height of the tree

Minimum, Maximum, Successor in a BST

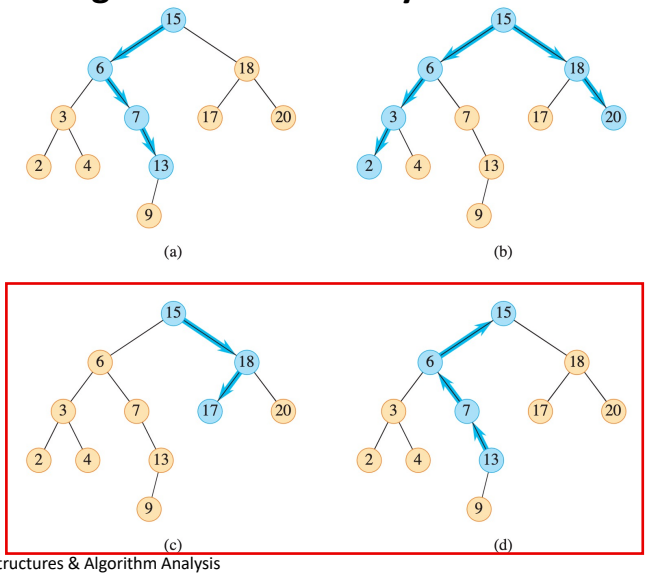
```
TREE-SUCCESSOR(x)
if x.right ≠ NIL
    return TREE-MINIMUM(x.right)   // leftmost node in right subtree
else // find the lowest ancestor of x whose left child is an ancestor of x
    y = x.p
    while y ≠ NIL and x == y.right
        x = y
        y = y.p
    return y
```

find 最小的更大的数字

Runtime: *O(h), h* the height of the tree

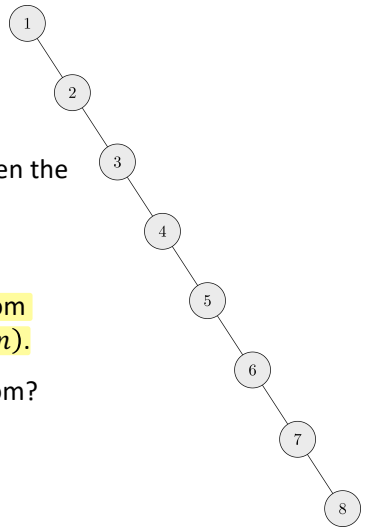


Searching in a BST: Summary



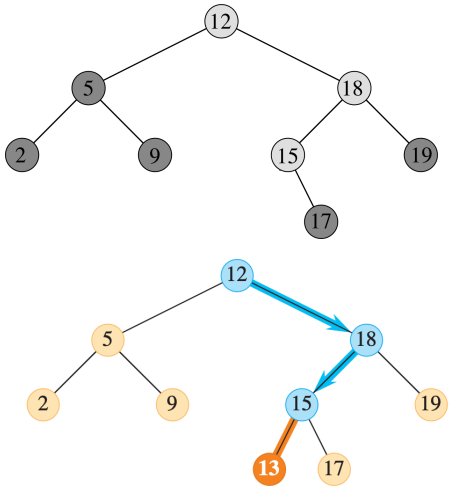
Searching in a BST: Worst case runtime

- BSTs can be **imbalanced** and even degenerate to a single path!
- Height can be as bad as *n-1,* e.g. when the input is sorted.
- So the **worst-case runtime is $\Theta(n)$.**
- If keys are inserted in uniform random order, the expected height is $O(\log n)$.
- Can we rely on our data being random? Such inputs might be very unlikely.
- We'll see **balanced trees** later on, guaranteeing a height of $O(\log n)$.



Insert(T,*z*)

Example: Insert(T,13)



Idea

Go down the tree like in Search to find where the new element needs to go.

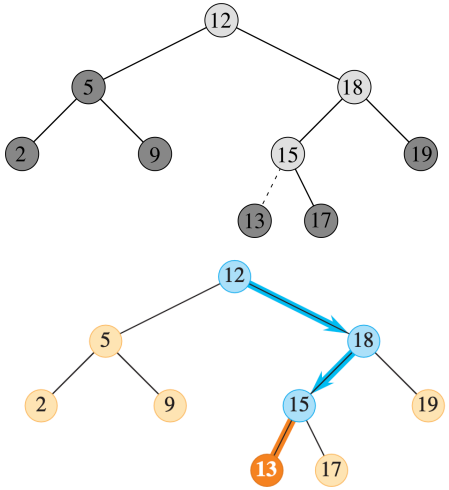
1. The search will end in NIL, hence we record the **search path** (e.g. 12, 18, 15, NIL).
2. Add the element as a left or right subtree to last non-NIL node.

Insert(T,z)

```
TREE-INSERT(T, z)
x = T.root
y = NIL
while x ≠ NIL
    y = x
    if z.key < x.key
        x = x.left
    else x = x.right
z.p = y
if y == NIL
    T.root = z
elseif z.key < y.key
    y.left = z
else y.right = z
```

说明树是空的

Example: Insert(T,13)

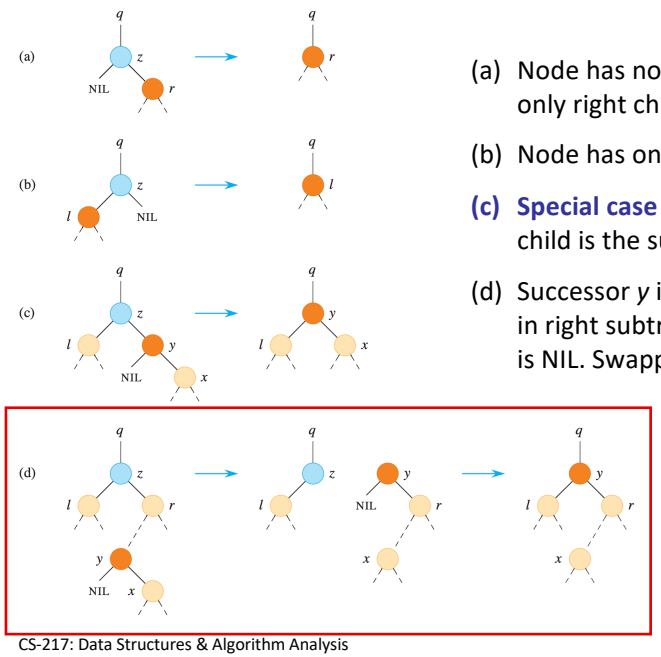


Runtime: *O(h), h* the height of the tree

Delete(T,z)

- **Idea:** Three cases

1. Easy when z is a **leaf** (delete z).
2. If **z has one child**, have the child replace z.
3. Otherwise, if **z has two children**, we can't leave a hole in the tree!
   - Solution: replace z with its **successor**.
   - z's successor is the minimum in the right subtree (this subtree exists since z has two children).
   - z's successor has no left child.
   - Hence we can swap it with z and then delete z.



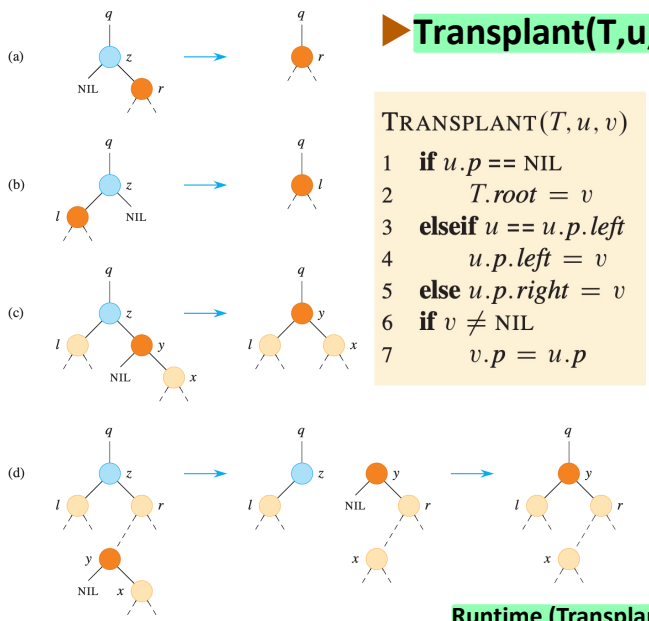
(a) Node has no children or only right child

(b) Node has only left child

(c) **Special case** where right child is the successor.

(d) Successor *y* is the minimum in right subtree; *y*'s left child is NIL. Swapping z and y.

Transplant(T,u,v)

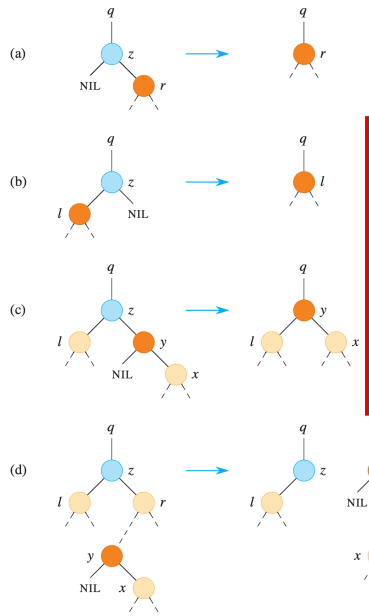



```
TRANSPLANT(T, u, v)
if u.p == NIL
    T.root = v
elseif u == u.p.left
    u.p.left = v
else u.p.right = v
if v ≠ NIL
    v.p = u.p
```

Replaces subtree rooted in u with a subtree rooted in v

Transplant does not update $v.left$ and $v.right$: this is the responsibility of the caller of ***Transplant!!***

Runtime (Transplant) ?
O(1)



Delete(T,z)

```
TREE-DELETE(T, z)
if z.left == NIL
    TRANSPLANT(T, z, z.right)          // replace z by its right child
elseif z.right == NIL
    TRANSPLANT(T, z, z.left)           // replace z by its left child
else y = TREE-MINIMUM(z.right)         // y is z's successor
    if y ≠ z.right                     // is y farther down the tree?
        TRANSPLANT(T, y, y.right)      // replace y by its right child
        y.right = z.right              // z's right child becomes
        y.right.p = y                  //     y's right child
    TRANSPLANT(T, z, y)                // replace z by its successor y
    y.left = z.left                    // and give z's left child to y,
    y.left.p = y                       //     which had no left child
```

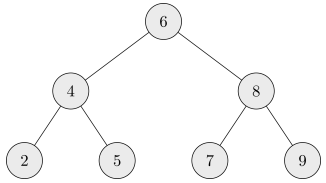
Runtime: *O(h), h* the height of the tree

Tree walks

- We can print out the keys of a BST by a **tree walk**:

PREORDER(x)	INORDER(x)	POSTORDER(x)
if $x \neq$ NIL **then**	**if** $x \neq$ NIL **then**	**if** $x \neq$ NIL **then**
print x.key	INORDER(x.left)	POSTORDER(x.left)
PREORDER(x.left)	print x.key	POSTORDER(x.right)
PREORDER(x.right)	INORDER(x.right)	print x.key

- Inorder tree walk outputs sorted sequence.



Inorder: 2, 4, 5, 6, 7, 8, 9
Preorder: 6, 4, 2, 5, 8, 7, 9
Postorder: 2, 5, 4, 7, 9, 8, 6

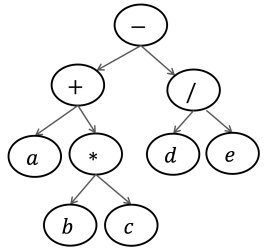
Tree walks: runtime

- **Theorem**: Inorder (Preorder/Postorder) tree walk of the root of an *n*-node tree takes time $\Theta(n)$.

PREORDER(x)	INORDER(x)	POSTORDER(x)
if $x \neq$ NIL **then**	**if** $x \neq$ NIL **then**	**if** $x \neq$ NIL **then**
print x.key	INORDER(x.left)	POSTORDER(x.left)
PREORDER(x.left)	print x.key	POSTORDER(x.right)
PREORDER(x.right)	INORDER(x.right)	print x.key

- Book gives a rather dull proof based on recurrences.
- A simpler proof:
  - Assign costs (time) for operations made at x to node *x*.
  - Cost at each node is $\Theta(1)$, and all costs are accounted for.
  - Sum of costs = runtime is $n \cdot \Theta(1) = \Theta(n)$.
- NB: This kind of argument is called **accounting method**.

Algebraic Expressions



- Algebraic expression with binary operators
- + - * /
- We can use a binary tree to represent it because the operations are binary
- Internal nodes: **operators**
- Leaves: **operands**
- (a+(b*c)) – (d/e)

- Inorder tree walk? **((a+(b*c))-(d/e))** Print **(** before left visit and **)** after right subtree visit
- Postorder tree walk? **a b c * + d e / -**
- Preorder tree walk? **- (+(a,*(b,c)),/(d,e))** add commas **,** after left visit

Inorder: infix expression; **Postorder:** postfix expression (stack);
Preorder: functional programming notation

▶Summary

- Binary trees have at most 2 children and can be defined recursively:
  - A tree is either empty or it contains a root and two subtrees (=trees).
  - Very useful for inductive proofs for trees.
- Binary search trees store data such that smaller keys are in the left subtree and larger keys are in the right subtree.
- BSTs of height h execute the following operations in time $O(h)$
  - **Searching, Minimum, Maximum, Successor**
  - **Insertion**
  - **Deletion**
- Binary search trees can be **imbalanced:** trees can degenerate to height $h = \Theta(n)$ **and worst-case time $\Theta(n)$** for many operations.
- Inorder/preorder/postorder walks output all elements in time $\Theta(n)$.